

Short Communication

Institute for Environmental Studies, Vrije Universiteit Amsterdam, The Netherlands

Autoregressive Conditional Heteroscedasticity in Daily Wind Speed Measurements

R. S. J. Tol

With 6 Figures

Received November 27, 1995

Summary

It is argued that the predictability of meteorological variables is not constant but shows regular variations. This is shown for the daily mean wind speeds and its meridional and zonal components at Shearwater, Canada, for the period 1963–1988. To capture this feature, a Generalised Auto Regressive Conditional Heteroscedastic model is proposed. In this model, the conditional variance of an observation depends linearly on the conditional variances of the previous observations and on the previous prediction errors. Here, conditional heteroscedasticity models are used which let the variance depend on previous prediction errors, in conjunction with an autoregressive model for the mean, using the Gamma distribution for the wind speed and the Normal distribution for its components. It is shown that these heteroscedastic models outperform their homoscedastic versions, and that heteroscedastic features are more clear in the wind speed component records.

1. Introduction

At some times weather prediction is easier than at other times. Periods of high quality forecasts alternate with periods of low quality forecasts. In order to improve the forecasts as well as the description and the stochastic generation of weather variables, it is therefore worthwhile to be able to recognise and model changes in the predictability. This note presents a class of models capable of doing so. The quality of a forecast (predictability) is inversely related to the variance

of the forecast error (volatility). The relationship between subsequent forecast errors is the core of the class of models discussed here; this relationship makes the models capable of dealing with the changes in predictability. This class of models are referred to as AutoRegressive Conditional Heteroscedasticity (ARCH) models, developed in financial econometrics to handle the changes in the volatility of stock and foreign exchange markets. In this note, (generalised) ARCH models are used to model the daily mean wind speeds at Shearwater, at the East Coast of Canada. In an companion paper (Tol, 1996), these models are used for the daily mean temperatures of De Bilt, The Netherlands. The models with conditionally changing variances are shown to outperform the models with constant variances (homoscedastic models).

Section 2 shows that the changes in predictability described above are indeed observed in the daily mean wind speeds at Shearwater. Section 3 is a short treaty on the theoretical background of (generalised) ARCH models, which model the variance of a process in analogue to autoregressive moving average (ARMA) models for a process' mean (Box and Jenkins, 1976). Section 4 uses a Gamma AR(2)-GARCH(1,1) model to describe the daily mean wind speeds. Section 5 presents a Normal AR(2)-GARCH(1,1) model

for the zonal and meridional components of the wind speeds. Section 6 concludes.

2. Motivation

The data-set used to motivate and exemplify the use of ARCH models in meteorology consists of the daily mean wind speed measurements at Shearwater, Canada, of the period September 1985 to February 1986. This period is chosen because this period is the one with the highest mean wind speed and reasonably homogeneous sample mean and variance on the whole sample of 1963–1988. Storm seasons are indicated by the year in which they end.

Figure 1 depicts the wind speed in the season of 1986 at Shearwater, W , as observed and as modelled by an autoregressive (AR) Gamma model of the second order:

$$W_t \sim \gamma(\alpha_t, \beta_t) \quad (1a)$$

with density

$$\gamma'(w; \alpha, \beta) = \frac{\alpha^\beta}{\Gamma(\beta)} e^{-\alpha w} w^{\beta-1} \quad (1b)$$

where α, β, w non-negative,

$$\Gamma(\beta) = \int_0^\infty y^{\beta-1} e^{-y} dy$$

and

$$\alpha_t = \frac{\mu_t}{\sigma_t^2} \quad \beta_t = \frac{\mu_t^2}{\sigma_t^2} \quad (1c)$$

with μ_t and σ_t the conditional mean and standard deviation of W at time t , following

$$\mu_t = C + \varphi_1 W_{t-1} + \varphi_2 W_{t-2} \quad \sigma_t^2 = \sigma^2 \quad (1d)$$

The Gamma distribution was chosen because it is defined on the positive real line. It is preferred over the Lognormal distribution in order to demonstrate the use of AR-GARCH models under non-normality. The parameters are estimated using numerical maximum likelihood, conditional on the two pre-sample observations. It appears that the model can only describe the data to some extent: the data contain more volatility than the model can capture. Additional AR-lags, moving average parameters or non-linear autoregression could not resolve this problem. The top rows of Table 1 contain the estimates of the parameters of the Gamma AR(2) model for the seasons of 1963 to 1988 inclusive (note that 1968 is missing because of missing observations). The results for the Gamma AR(0) model (assuming a constant mean), and the Gamma AR(2)-GARCH(1, 1) and Gamma AR(2)-GARCH(2, 2) models (to be explained below) are also taken up. Comparing the

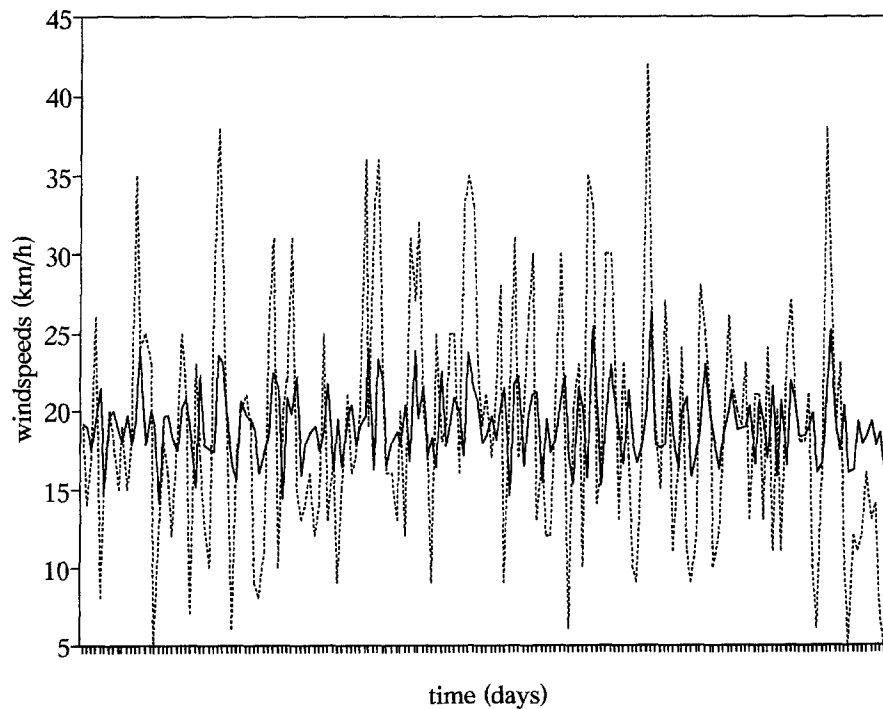


Fig. 1. The daily mean wind speeds at Shearwater, Canada, of the storm season of 1986 as observed (dashed line) and as modelled by a Gamma AR(2) process (solid line)

1969		1970		1971		1972		1973	
compound	wind speed	16.062	16.067	16.202	16.203	17.519	17.594	17.854	17.809
μ	0.000	0.000	0.177	0.198	0.208	0.000	0.210	0.201	0.202
σ	0.362	0.362	0.130	0.138	0.208	0.000	0.042	0.027	0.028
λ	-0.024	-0.024	-0.130	-0.138	-0.208	0.000	-0.042	-0.027	-0.028
μ	0.000	0.000	0.130	0.138	0.208	0.000	0.042	0.027	0.028
σ	0.362	0.362	0.130	0.138	0.208	0.000	0.042	0.027	0.028
λ	-0.024	-0.024	-0.130	-0.138	-0.208	0.000	-0.042	-0.027	-0.028
μ	0.000	0.000	0.130	0.138	0.208	0.000	0.042	0.027	0.028
σ	0.362	0.362	0.130	0.138	0.208	0.000	0.042	0.027	0.028
λ	-0.024	-0.024	-0.130	-0.138	-0.208	0.000	-0.042	-0.027	-0.028
μ	0.000	0.000	0.130	0.138	0.208	0.000	0.042	0.027	0.028
σ	0.362	0.362	0.130	0.138	0.208	0.000	0.042	0.027	0.028
λ	-0.024	-0.024	-0.130	-0.138	-0.208	0.000	-0.042	-0.027	-0.028
μ	0.000	0.000	0.130	0.138	0.208	0.000	0.042	0.027	0.028
σ	0.362	0.362	0.130	0.138	0.208	0.000	0.042	0.027	0.028
λ	-0.024	-0.024	-0.130	-0.138	-0.208	0.000	-0.042	-0.027	-0.028
μ	0.000	0.000	0.130	0.138	0.208	0.000	0.042	0.027	0.028
σ	0.362	0.362	0.130	0.138	0.208	0.000	0.042	0.027	0.028
λ	-0.024	-0.024	-0.130	-0.138	-0.208	0.000	-0.042	-0.027	-0.028
μ	0.000	0.000	0.130	0.138	0.208	0.000	0.042	0.027	0.028
σ	0.362	0.362	0.130	0.138	0.208	0.000	0.042	0.027	0.028
λ	-0.024	-0.024	-0.130	-0.138	-0.208	0.000	-0.042	-0.027	-0.028
μ	0.000	0.000	0.130	0.138	0.208	0.000	0.042	0.027	0.028
σ	0.362	0.362	0.130	0.138	0.208	0.000	0.042	0.027	0.028
λ	-0.024	-0.024	-0.130	-0.138	-0.208	0.000	-0.042	-0.027	-0.028
μ	0.000	0.000	0.130	0.138	0.208	0.000	0.042	0.027	0.028
σ	0.362	0.362	0.130	0.138	0.208	0.000	0.042	0.027	0.028
λ	-0.024	-0.024	-0.130	-0.138	-0.208	0.000	-0.042	-0.027	-0.028
μ	0.000	0.000	0.130	0.138	0.208	0.000	0.042	0.027	0.028
σ	0.362	0.362	0.130	0.138	0.208	0.000	0.042	0.027	0.028
λ	-0.024	-0.024	-0.130	-0.138	-0.208	0.000	-0.042	-0.027	-0.028
μ	0.000	0.000	0.130	0.138	0.208	0.000	0.042	0.027	0.028
σ	0.362	0.362	0.130	0.138	0.208	0.000	0.042	0.027	0.028
λ	-0.024	-0.024	-0.130	-0.138	-0.208	0.000	-0.042	-0.027	-0.028
μ	0.000	0.							

Table 1d. Results for the Daily Mean Wind Speeds at Shearwater, Canada, 1979–1983

1979		1980		1981		1982		1983											
compound wind speed																			
μ	18.732	18.732	17.483	17.479	17.623	17.149	16.777	16.789	16.855	16.886	18.492	18.576	18.815	18.766	16.123	16.163	16.132		
μ_1	0.000	0.334	0.000	0.343	0.337	0.328	0.328	0.163	0.161	0.164	0.000	0.295	0.288	0.320	0.117	0.126	0.140		
μ_2	0.000	0.009	0.029	0.013	0.014	0.014	0.014	0.071	0.069	0.072	0.058	0.031	0.034	0.036	0.001	0.002	0.012		
μ_3	50.092	42.448	43.773	65.109	54.013	53.925	75.430	72.069	72.720	76.253	60.705	54.105	55.694	54.557	43.323	42.581	42.606		
λ_1	0.000	0.000	0.042	0.000	-0.030	-0.034	0.000	0.000	-0.022	0.006	0.000	0.000	-0.047	-0.079	0.000	0.026	-0.026		
λ_2	0.000	0.000	0.090	0.000	0.000	-0.090	0.000	0.000	0.000	0.086	0.000	0.000	0.000	0.000	0.000	0.000	0.000		
λ_3	0.000	0.000	-0.719	0.812	0.960	0.616	0.000	0.000	-0.452	0.837	0.000	0.000	0.978	0.990	0.000	0.531	-0.236		
k_1	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000		
k_2	0.000	0.000	-0.121	0.000	0.000	-0.580	0.000	0.000	0.000	0.017	0.000	0.000	0.000	-0.702	0.000	0.824	-0.236		
k_3	0.000	0.000	0.000	0.000	0.000	-0.580	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000		
LL	-593.773	-581.536	-581.230	-601.745	-601.078	-600.403	-623.172	-619.734	-619.701	-618.531	-609.419	-598.826	-597.542	-596.435	-580.210	-578.639	-578.337	-576.436	
meridional component																			
μ	2.821	2.822	2.773	2.940	3.185	3.191	2.145	2.091	2.200	2.374	1.843	1.827	1.409	1.049	2.761	2.759	2.832	2.861	
μ_1	0.000	0.305	0.318	0.312	0.342	0.343	0.000	0.376	0.397	0.422	0.000	0.208	0.225	0.225	0.000	0.306	0.304	0.306	
μ_2	0.000	0.108	0.146	0.159	-0.013	-0.023	0.000	-0.059	-0.065	-0.079	0.000	0.000	0.030	0.017	0.000	-0.079	-0.071	-0.076	
μ_3	190.964	166.632	169.007	188.707	167.641	165.953	173.301	151.012	152.806	153.448	127.447	121.991	125.338	133.769	158.596	144.896	144.794	144.807	
λ_1	0.000	0.000	0.128	0.063	0.243	0.258	0.000	0.000	0.121	0.056	0.000	0.000	0.251	0.228	0.000	0.000	0.141	0.141	
λ_2	0.000	0.000	0.000	0.121	0.000	-0.041	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.030	0.000	0.000	0.074	0.074	
λ_3	0.000	0.000	0.702	0.577	0.254	0.256	0.000	0.000	0.658	0.618	0.000	0.000	0.468	-0.671	0.000	0.000	-0.554	-0.554	
k_1	0.000	0.000	-0.017	0.000	0.000	0.023	0.000	0.000	0.000	0.000	0.000	0.000	0.000	-0.347	0.000	0.000	-0.357	-0.357	
k_2	0.000	0.000	0.000	0.000	0.000	-0.539	0.261	0.539	0.261	-0.539	0.376	0.523	0.370	-0.515	0.689	0.523	0.370	0.523	
k_3	0.000	0.000	-0.000	0.000	0.000	0.000	-0.550	0.875	-0.550	0.875	-0.536	0.840	-0.536	0.840	-0.515	0.689	-0.536	0.840	
LL	-559.582	-547.366	-545.518	-544.924	-550.208	-543.361	-559.261	-539.261	-539.261	-536.840	-536.376	-523.370	-519.453	-516.274	-515.689	-542.939	-534.855	-533.931	-533.660
zonal component																			
μ	-4.148	-4.069	-3.399	-3.350	-3.797	-3.782	-4.855	-4.873	-5.646	-5.807	-7.330	-7.286	-6.807	-5.565	0.200	0.171	0.282	0.272	
μ_1	0.000	0.279	0.279	0.277	0.279	0.270	0.000	0.159	0.169	0.128	0.000	0.212	0.223	0.252	0.000	0.289	0.335	0.336	
μ_2	0.000	0.008	0.056	0.055	-0.002	-0.024	0.000	-0.127	-0.132	-0.021	0.000	0.052	0.066	0.016	0.000	-0.061	-0.052	-0.051	
μ_3	180.891	166.905	171.991	172.831	158.621	162.279	148.879	143.209	144.927	145.700	218.335	206.828	206.609	183.260	137.631	126.919	127.053	127.053	
λ_1	0.000	0.000	0.195	0.214	0.245	0.239	0.000	0.000	0.223	0.213	0.000	0.000	0.077	0.104	0.000	0.110	0.112	0.112	
λ_2	0.000	0.000	0.000	-0.017	0.000	0.066	0.000	0.000	0.000	-0.065	0.000	0.000	0.000	0.020	0.000	0.000	0.000	-0.010	
λ_3	0.000	0.000	0.573	0.570	0.377	0.386	0.000	0.000	-0.357	0.365	0.000	0.000	-0.667	-0.789	0.000	-0.278	-0.176	-0.176	
k_1	0.000	0.000	-0.002	0.000	0.000	-0.105	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.232	0.000	0.000	0.000	0.075	

[illegible]

A closer look at Fig. 1 reveals there are periods in which the model performs better than in others, and that there seems to be some pattern in this. Figure 2 illustrates this more clearly: the dashed line depicts the absolute residuals (observed minus modelled data). The straight solid depicts the estimated standard error. It is clear that large prediction errors have some tendency to be followed by small ones and vice versa. For some periods the forecast errors tend to be smaller than the standard error, for other periods the errors tend to be larger. The other solid line is the conditional variance according to the GARCH(1,1) model, which takes into account that large errors are likely to be followed by small errors; this is explained in the next section.

The central concept of ARCH modelling is that periods of large and small forecast errors alternate. In this case, the observation of one large (small) prediction error yields the information that the next period is more likely to have also a large (small) prediction error than to have a small (large) one. This is an analogue to autoregression in the mean, in which large (small) observations are likely to be succeeded by large (small) ones. A logical choice is then to condition the model's variance on the forecast error(s) of the previous periods. With μ denoting the expected wind speed and with σ^2 the variance of the wind speed, this leads to:

where F is a probability distribution (e.g., a Gamma), and with

This is the original ARCH(p) model due to Engle (1982). The second part of Eq. (1d) can readily be replaced with (2b). Note that then each observation W is modelled conditional on its past in both mean and variance. The unconditional distribution of W has fatter tails than the conditional distribution (Bollerslev et al., 1992). In this model,

the variance depends on a constant, α , and the weighted, by λ_i , and truncated, at lag p , sum of the lagged squared prediction errors, $(W_{t-i} - \mu_{t-i})^2$. Thus, a large (small) prediction error results in an increase (decrease) of the conditional variance, representing a loss (gain) in predictability; in fact, every observation updates the variance. Notice that the expectation of the squared residual at time t equals the conditional variance at time t . The parameters of the ARCH model are subject to a number of restrictions. Obviously, all conditional variances have to be positive. Also, for most applications it is preferable to have stationary conditional variances (and so a bounded unconditional variance), i.e., the roots of the polynomial should lie outside the unit circle.

In empirical applications of ARCH models, a relatively long lag (i.e., a large p) is often called for, and to avoid problems with negative variance parameter estimates a fixed lag structure is typically imposed. To overcome these problems, Bollerslev (1986) generalises Engle's ARCH model by also introducing lagged conditional variances. Again (without loss of generality) assuming normality, the GARCH(p, q) looks like:

$$W_t \sim F(\mu_t, \sigma_t^2) \quad (3a)$$

with

$$\sigma_t^2 = \alpha + \sum_{i=1}^p \lambda_i (W_{t-i} - \mu_{t-i})^2 + \sum_{i=1}^q \kappa_i \sigma_{t-i}^2 \quad (3b)$$

Thus, the conditional variance of W_t equals a constant, α , plus the weighted, by λ_i , and truncated, at lag p , sum of past squared prediction errors, $(W_{t-i} - \mu_{t-i})^2$, and the weighted, by κ_i , and truncated, at lag q , sum of past conditional variances, σ_{t-i}^2 . Again, restrictions on α , λ_i , κ_i are needed in order to keep the conditional variances non-negative and the unconditional variance bounded.

Analogue to the relationship between the autoregressive (AR) and moving average (MA) parameters in an ARMA model, the GARCH model can be seen as a parsimonious approximation of long-lagged ARCH model (in fact, GARCH corresponds to a restricted ARCH(∞) model). Taylor (1986), who developed the GARCH model independently, indeed coined the name ARMACH.

The publication of the first articles on ARCH models, and their successful applications, especially to financial series, has led to rapidly

expanding bulk of literature on applications, extensions and theoretical background of these models. This has been reviewed in Bollerslev et al. (1992). The extensions include non-linear, threshold and asymmetrical update formulae, ARCH-in-mean models, structural models, multivariate models, and non-stationary models for the variance. The last extension led Engle and Bollerslev (1993) to propose the concept of co-persistence, according to which the non-stationary variances of two variables are tied together in the long run. In this note, we restrict the attention to the basic forms.

4. Daily Wind Speed Volatility

A GARCH(1,1) model is applied to the conditional variance of the storm season 1986 wind speeds at Shearwater. Its parameters are again obtained by the method of maximum likelihood, conditional upon the first two observations. The expectation of the first observation is set at the overall (estimated) mean, the first prediction error is set at the difference between the first observation and its prediction error; the variance of the first observation is set at the overall (estimated) mean variance. The moments of the second observation assume that the presample observation equals the overall mean, the presample prediction error is zero, and the presample variance equals the overall mean variance. The mean wind speeds are still modelled by a Gamma AR(2) process, see Eq. (1); note that its parameters are re-estimated. The solid line in Fig. 2 depicts the modelled GARCH(1,1) conditional standard deviations, the straight solid line represents the standard deviation of homoscedastic model. Note that the heteroscedastic model allows the base-line variance to be lower than the homoscedastic model, which implies an additional gain in descriptive power. The loglikelihood of the AR(2) model is about 5 points lower than the loglikelihood of the AR(2)-GARCH(1,1) model, which justifies the two additional parameters according to Akaike's Information Criterion.

The upper rows of Table 1 contain the estimated parameters for the Gamma AR(2)-GARCH(2,2) model for the Shearwater wind speeds of 1963 to 1988 inclusive, as well as the loglikelihoods; from right to left, more restrictions are imposed: the right most column presents the

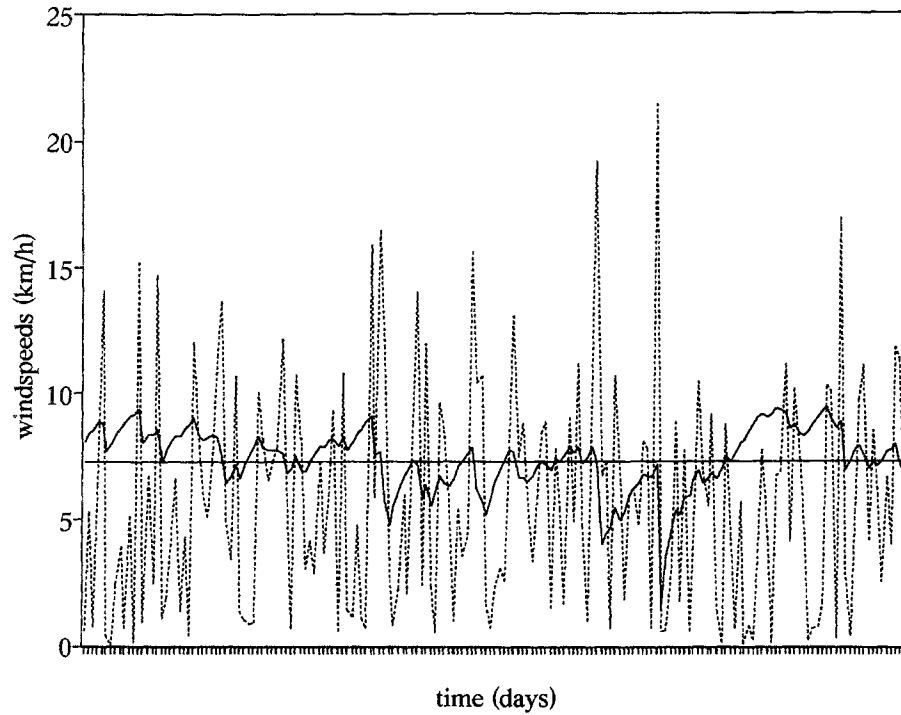


Fig. 2. The volatility of the daily mean wind speeds at Shearwater, Canada, of the storm season of 1986 as observed (dashed line – absolute residuals of the AR(2) model) and as modelled by the homoscedastic Gamma AR(2) model (straight solid line – standard deviations) and by the heteroscedastic Gamma AR(2)-GARCH(1,1) model (solid line – standard deviations)

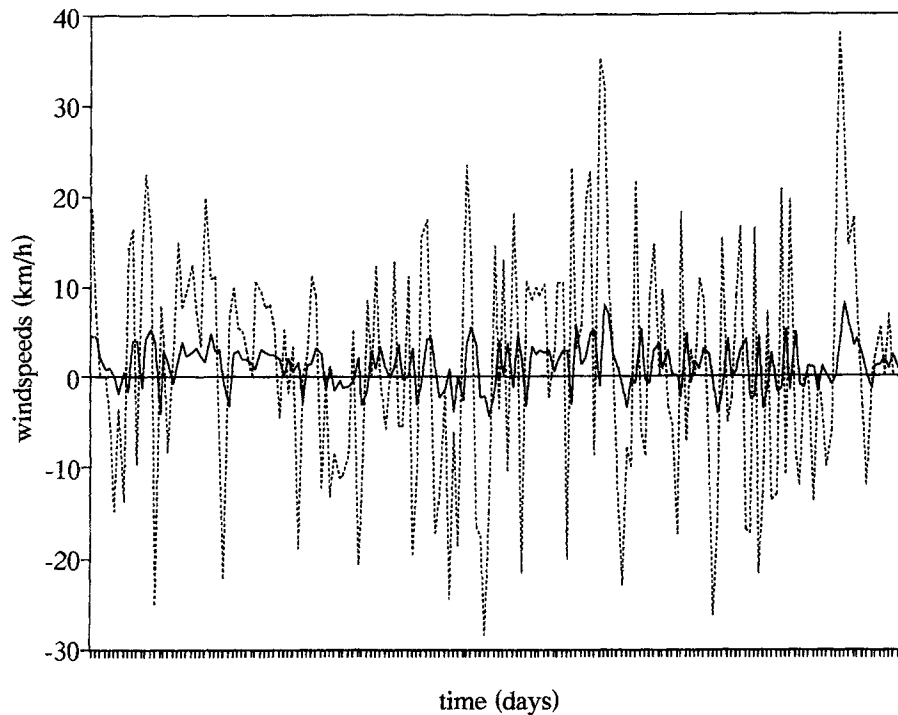


Fig. 3. The daily mean meridional wind speeds at Shearwater, Canada, of the storm season of 1986 as observed (dashed line) and as modelled by a Normal AR(2) process (solid line)

outcomes for the most general model – AR(2)-GARCH(2, 2) – the left most the most restricted – AR(0)-GARCH(0,0). Overall, the AR(2) model outperforms the constant mean model, and the heteroscedastic models outperform the homoscedastic ones; in most cases, the GARCH(1,1)

model is sufficient but in some years, e.g., 1975, the more elaborate GARCH(2, 2) structure is called for. On the whole, Table 1 provides evidence that weak heteroscedasticity is present in daily wind speed measurements, and that it pays to take this into account in the statistical models.

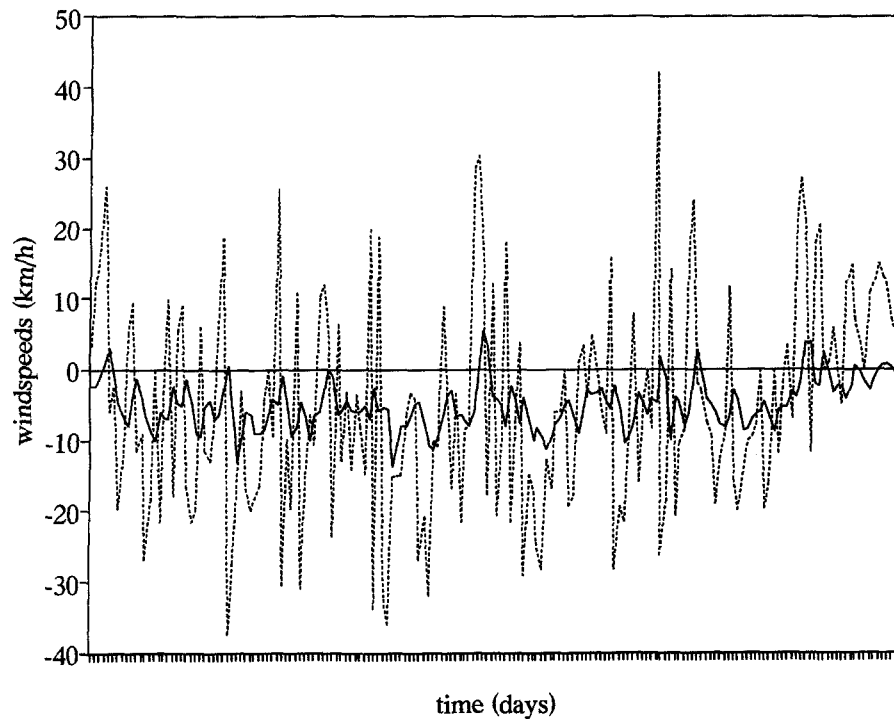


Fig. 4. The daily mean zonal wind speeds at Shearwater, Canada, of the storm season of 1986 as observed (dashed line) and as modelled by a Normal AR(2) process (solid line)

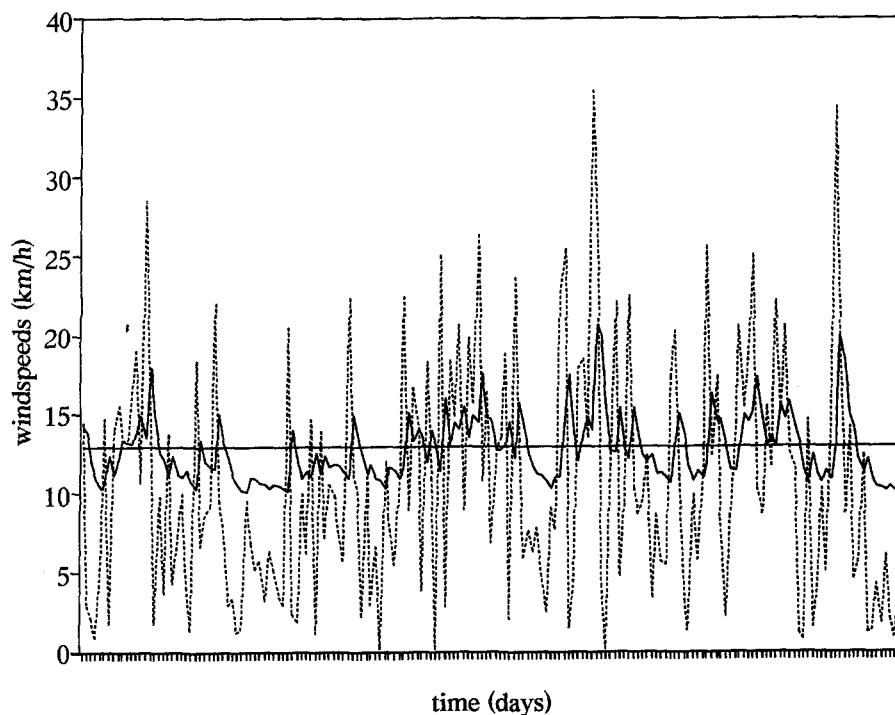


Fig. 5. The volatility of the daily mean meridional wind speeds at Shearwater, Canada, of the storm season of 1986 as observed (dashed line – absolute residuals of the AR(2) model) and as modelled by the homoscedastic Normal AR(2) model (straight solid line – standard deviations) and by the heteroscedastic Normal AR(2)-GARCH(1,1) model (solid line – standard deviations)

5. Daily Wind Speed Component Volatility

It is concluded from the previous section that taking up heteroscedasticity in the statistical model to describe daily wind speeds does pay. However, wind speed is defined as the square root of the sum of the squares of meridional and zonal

wind speeds, which are measured. In this section, independent Normal AR(2)-GARCH(2,2) models are used for the meridional and zonal winds. Possible dependencies between the mean and variance of the wind components are neglected. Figures 3 and 4 depict the observed and modelled meridional and zonal wind speeds, respectively,

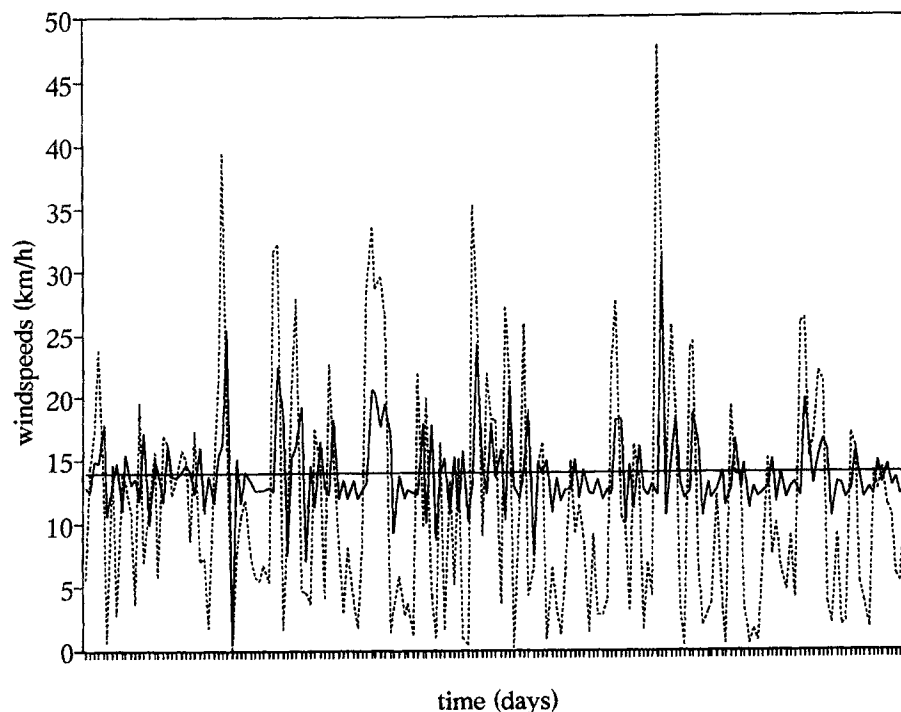


Fig. 6. The volatility of the daily mean zonal wind speeds at Shearwater, Canada, of the storm season of 1986 as observed (dashed line – absolute residuals of the AR(2) model) and as modelled by the homoscedastic Normal AR(2) model (straight solid line – standard deviations) and by the heteroscedastic Normal AR(2)-GARCH(1,1) model (solid line – standard deviations)

for the storm season of 1986. Again, the observations are much more volatile than the model, which reflects itself in the large variances; this is common for the investigated seasons (see Table 1).

Figures 5 and 6 depict the residual and modelled standard deviations for the meridional and zonal wind speeds of 1986. There are clear clusters of high and low forecast errors. These figures illustrate the power of GARCH(1,1) models more clearly than the corresponding Fig. 2. In Fig. 5, the standard deviation of the meridional wind speeds rises and falls and the modelled standard deviation follows, sometimes falling below the standard deviation in the homoscedastic model, and sometimes rising above it. In Fig. 6, this is even more pronounced: the zonal wind speeds' standard deviation behaves, at first sight, erratically but the GARCH(1,1) is capable of following and providing a reasonable, time-dependent estimate of the forecast error – the loglikelihood jumps 20 points up (cf. Table 1). The parameter estimates for the other storms seasons can be found in Table 1. On the whole, the GARCH(1,1) model adds to the description of the zonal and meridional wind speeds although sometimes the more elaborate GARCH(2,2) model is required.

6. Conclusion

In this note, it is argued that the volatility of wind speed records tends to cluster, i.e., the predictability of the weather is not constant but shows some systematic variation. This feature can be captured by an autoregressive conditional heteroscedastic (ARCH) model for the conditional variances of the data. Here, the GARCH(1,1) model is employed. In this model, the conditional variance at time t depends on the prediction error and the conditional variance at time $t - 1$. It is shown that the proposed heteroscedastic models outperform the homoscedastic ones for wind speed and wind speed components at Shearwater, Canada. A companion paper reaches a similar conclusion for daily summer and winter temperatures at De Bilt, The Netherlands (Tol, 1996).

(Generalised) ARCH models can be used to support meteorological services in assessing forecast errors of daily weather predictions. The resulting conditional variances can also be used to identify the stability of the weather, or to compare this stability for different climates.

Future research could study whether other meteorological records exhibit ARCH features, and whether artificial weather corresponds better to

the observed if ARCH features are built into stochastic weather generators.

Acknowledgements

This author hereby expresses his gratitude to T. A. Buishand, an anonymous referee and the managing editor for constructive comments, M. C. Beefink for the data, and H. R. Merkus for the software.

References

- Bollerslev, T., 1986: Generalized autoregressive conditional heteroscedasticity. *J. Econometrics*, **31**, 307–327.
- Bollerslev, Y., Chou, R. Y., Kroner, K. F., 1992: ARCH modelling in finance. *J. Econometrics*, **52**, 5–59.
- Box, G. E. P., Jenkins, G. M., 1976: *Times Series Analysis – Forecasting and Control*, 2nd edn. Oakland: Holden-Day.
- Engle, R. F., 1982: Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, **50**, 987–1007.
- Engle, R. F., Bollerslev, T., 1993: Common persistence in conditional variance. *Econometrica*, **61**, 167–186.
- Taylor, S. J., 1986: *Modelling Financial Time Series*. Chichester: Wiley.
- Tol, R. S. J., 1996: Autoregressive conditional heteroscedasticity in daily mean temperatures. *Environmetrics* **7**(1), 67–76.
- Authors' address: Richard S. J. Tol, Institute for Environmental Studies, Vrije Universiteit, De Boelelaan 1115, NL-1081 HV Amsterdam, The Netherlands.